

Generalized Linear Model: Assignment 3

Student ID:

Name:

1. Check whether each statement is true:

- (1) X, Y, Z : mutual independence $\Rightarrow$ joint independence
- (2) $Y \perp (X, Z)$ (joint independence) $\Rightarrow X \perp Y|Z$ and $Y \perp Z|X$
- (3) $X \perp Y|Z$ and $Y \perp Z|X \Rightarrow X \perp Y$ and $Y \perp Z$

2. Prove that if $\theta_{11(1)} = \dots = \theta_{11(K)}$, $\frac{1}{4} \log \theta_{11(K)} = \lambda_{11}^{XY}$

3. Check whether each statement is true:

- (1) If $X \perp Y|Z$ for three-way contingency table, it is collapsible
- (2) The homogeneous association, (XY, XZ, YZ) , is collapsible

4. Consider the following Poisson log-linear model for an $I \times J$ contingency table:

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y$$

Prove the following:

(1) $\log \frac{\pi_{a+}}{\pi_{b+}} = \lambda_a^X - \lambda_b^X$

(2) If all $\lambda_j^Y = 0$, then $\pi_{+j} = \frac{1}{J}$

5. In a certain area, the average number of injury accidents is 1, and the average number of non-injury accidents is 3. If a total of 10 accidents occurred, what is the probability that 2 of them were injury accidents?

6. Suppose, on average, there is 1 bug per 500 lines of code. If a programmer writes five programs, each with 300 lines, what is the probability that 2 or fewer bugs occur in total?

7. Suppose $X_1 \sim \text{Poisson}(\lambda_1), \dots, X_k \sim \text{Poisson}(\lambda_k)$ are independent random variables. Then, show the conditional distribution follows a multinomial distribution:

$$P(Y_1 = y_1, Y_2 = y_2 | Y_1 + \dots + Y_k = n)$$

8. Let $X \sim U(0, \theta)$, then draw then the likelihood function of θ is when the observed value of X is x

9. Suppose $Y \sim B(n, \pi)$, where n takes values 2 or 3, and π takes values $\frac{1}{2}$ or $\frac{1}{3}$. In this case, for each combination of n and π , determine the maximum likelihood estimate

10. If $Y \sim B(n, \pi)$ and we observe $Y = y$ with known n , find the MLE of π

11. If $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, 1)$, find the MLE of μ .

12. If $Y_1, \dots, Y_n \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\lambda)$, Find the MLE of $P(Y_1 = 0)$.

13. A survey was conducted on whether to legalize abortion for pregnant women.

$$H_0: \pi = 0.5 \text{ vs } H_1: \pi < 0.5$$

At the University of Chicago, among 893 respondents, 400 were in favor and 493 were opposed. What conclusion can be drawn?

14. In a public opinion survey evaluating the local government system, out of 500 respondents, 165 gave a positive response. Construct a 95% confidence interval for the proportion of people in the entire population who have a positive opinion about the local government system. $p = \frac{165}{500} = 0.33$, $s.e.(p) = 0.012$

15. The previous highest cure rate is 0.3. A new anticancer drug was tested on $n = 13$ patients, and $y = 7$ were cured (≥ 5 -year survival). Test whether the new drug is more effective than the old one. Compute both the p-value and the mid-p value.

y	$P(y)$	P -value	Mid P -value
0	0.001	1.000	0.9995
1	0.010	0.999	0.994
2	0.044	0.989	0.967
3	0.117	0.945	0.887
4	0.205	0.828	0.726
5	0.246	0.623	0.500
6	0.205	0.377	0.274
7	0.117	0.172	0.113
8	0.044	0.055	0.033
9	0.010	0.011	0.006
10	0.001	0.001	0.0005

16. Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, Find the Wald, Score, and Likelihood Ratio test statistics for the hypotheses: $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$, with σ^2 known.